Journal of Biomedical Informatics 170 (2025) 104888

Contents lists available at ScienceDirect

Journal of Biomedical Informatics

journal homepage: www.elsevier.com/locate/yjbin

Original Research

WoundcareVQA: A multilingual visual question answering benchmark dataset for wound care☆

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ARTICLE INFO

Keywords:
Visual question answering
Consumer health queries
Multimodal
Multilingual
Wound care

ABSTRACT

Objective: Introduce the task of wound care multimodal multilingual visual question answering, provide baseline performances, and identify areas of future study.
Methods: A dataset of wound care multimodal multilingual visual question answering (VQA) was created using consumer health questions asked online. Practicing US medical doctors were tasked with providing metadata and expert responses labels. Several instruct-enabled, multilingual visual question answering models (GPT-4o, Gemini-1.5-Pro, and Qwen-VL) were tested to benchmark performances. Finally, automatic evaluations were tested against domain expert response ratings.
Results: A multilingual dataset of 477 wound care cases, 768 responses, 748 images, 3k structured data labels, 1362 translation instances, and 10k judgments was constructed (https://osf.io/xsj5u/). Metadata scores ranged from 0.32–0.78 accuracy depending on classification type; response generation performances 0.06 BLEU, 0.66 BERTScore, 0.45 ROUGE-L in English and 0.12 BLEU, 0.69 BERTScore, and 0.50 ROUGE-L in Chinese.
Conclusion: We construct and explore the tasks of multimodal, multilingual VQA. We hope the work here can inspire further research in wound care metadata classification, VQA response generation, and open response automatic evaluation.

1. Introduction

Remote clinical care, via patient portals and remote visits, when appropriate, has been shown to reduce cost, increase access, and improve time-flexibility experiences for both patients and providers [1]. Before the COVID-19 pandemic, remote patient care was an emerging trend; however, it became essential, strengthened, and expanded due to the restrictions imposed by quarantine and social distancing measures, with the support of advancements in technology. Six years later, remote care has become ubiquitous, with an estimated 157% increase in patient messages compared to pre-pandemic [2]. Despite the advantages of this modern convenience, one major disadvantage of asynchronous remote care delivery is the dramatic workload increase for providers of, often uncompensated and unscheduled workload, requiring personal after-hour work, contributing to physician burnout [2,3]. Besides providing patient care in meetings, they are now additionally confronted with a “never-ending” inbox [2]. Automatic models, which consume patient queries and output response drafts, can provide suggestions to expedite throughput, alleviating provider burden.

Specifically, in this work, mirroring electronic messages sent by patients, we explore the task of multimodal text and image consumer health query response generation for the domain of wound care, in English and Chinese (Fig. 1). We provide the following contributions: (a) a new multimodal multilingual wound care dataset for Woundcare Visual Question Answering (WoundcareVQA) (https://osf.io/xsj5u/), (b) baseline model performances for metadata classification and response generation, critically including human domain expert evaluation, and (c) analysis on state-of-the-art model performances and directions for task future work and challenges. We hope the work created here may inspire future efforts on the development and evaluation of response generation systems that can assist healthcare providers in managing wound care-related questions more efficiently (see Table 1).

☆ This article is part of a Special issue entitled: ‘Multimodal LLM’ published in Journal of Biomedical Informatics.

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https://doi.org/10.1016/j.jbi.2025.104888
Received 4 February 2025; Received in revised form 17 July 2025; Accepted 28 July 2025
Available online 29 August 2025

Table 1
Statement of significance.

Problem or Issue: Remote clinical care (e.g. via electronic messaging), has opportunities to expand care and reduce cost; however it is at the expense of increased clinical workload. To alleviate this burden, we study the task of automatically generating responses to patient multimodal text and image queries for the wound care domain.
What is Already Known: Multimodal question-answering with multiple images, a textual context, and a question as input and a free response textual output, is a scarcely studied task. Previous work was on dermatology in a multilingual context, with BLEU 12.85, 7.05, 1.35 for English, Chinese, and Spanish respectively.
What this Paper Adds: The paper contributes a new benchmark on Woundcare Open-response Visual Question Answering dataset (https://osf.io/xsj5u/) and baselines using state-of-the-art models.
Who would benefit from the new knowledge in this paper: This work would be of interest to scientists developing models for visual question answering, health technologists incorporating technology to improve clinical workflows, and medical professionals wishing to understand AI model utility.

QUERY

求解啊 这怎么弄啊
Seeking for the advice! What should I do?

一个半月了，不消肿 ，是摔得，伤口特别深，有半厘米左右，伤口的周围红肿，一按有小坑，而且半天不回去，跟水肿了似得，怎么办啊？
It's been a month and a half, and the swelling hasn't gone down. It was a fall, and the wound was particularly deep, about half a centimeter. The area around the wound is red and swollen. When pressed, it forms a small pit that takes a long time to go back, looking like edema. What should I do?

RESPONSE

伤口迟迟未能愈合、伤口周边有红肿，很可能是感染。建议到紧急护理诊所拿取抗生素服用。
Delayed wound healing and surrounding redness and swelling of the area around the wound is suspicious for infection. Recommend evaluation in urgent care for antibiotics.

METADATA


```
{
    "anatomic_locations": ["forearm"],
    "wound_type": "traumatic",
    "wound_thickness": "stage_II",
    "tissue_color": "red_moist",
    "drainage_amount": "scant",
    "drainage_type": "serosanguinous",
    "infection": "infected”
}
```


Fig. 1. Example input multimodal query and response and metadata Labels.

2. Related work

2.1. Patient portal message classification, consumer health question-answering, and dialog generation for text

Prior work in patient portal message text processing in the United States has primarily centered around message classification. Specifically, several works explore categorizing messages to topics such as medical (e.g. problems, tests), social (e.g. expressing gratitude), logistical (e.g. medical records) [4–6]. This is useful for triaging messages by urgency or directing messages to appropriate resources (e.g. administrators for scheduling, medical doctors for clinical problems).

Outside a hospital patient portal setting, several efforts have explored textual dialog generation of entire patient-provider online conversations. Many primarily use Chinese datasets from patient-provider platforms such as Haodf.com, ChunYu YiSheng [7,8]. However, the MedDialog dataset also includes an English-language segment derived from service platforms like iclinic.com and healthcaremagic.com [7]. These studies primarily focus on the dialog generation task and measure natural language generation for each turn of a doctor given the entire conversation history using BLEU scores. Often performances are in lower ranges, reported 0.018 and 0.2399 BLEU for MedDialog and ChunYu, highlighting the difficulty of the task. Several United States National Library of Medicine (NLM) experiments assess textual single-turn questions and answers posed to United States National Institutes of Health (NIH) websites, including work on categorizing and semantically labeling consumer health queries [9], answering consumer health questions using retrieval approaches [10], and question entailment-based methods [11].

2.2. Wound image classification

Wound image classification has primarily focused on image classification with varying emphasis on specific subdomains and schemas. For example, some studies distinguish between burn and pressure ulcers [12], while others adopt more detailed taxonomies that incorporate various wound categories, such as diabetic or venous stasis ulcers [13]. Our analysis revealed that classification taxonomies are not universally standardized, with overlapping schema categories often determined by the specific areas of interest, such as ischemic diabetic foot wounds [14], leprosy [15], or post-operative wounds [16]. Different from other work, the authors of [17,18] integrated both location metadata and image data in their wound type classification (e.g., diabetic, venous).

2.3. Visual question answering

The majority of prior work in visual question answering (VQA) are artificially created using online images as prompts for generating questions and answers from humans [19,20]. Answers are typically 1–3 words, and in which performance can be measured using accuracy. Medical VQA datasets have been created similarly in the areas of radiology [21–24], pathology [25], and gastrointestinal images [26,27]. Unlike these prior work, our data includes a variable set of naturally-created questions, additional text context, and long answers.

One dataset similar to this work, in the general domain, is the VQAOnline dataset [28], which contains questions (accompanied by images) and answers sourced from authentic online interactions on platforms such as Stack Exchange. The highest reported performance metrics for this dataset include a ROUGE-L score of 0.16 and a BERTScore of 0.76. In the medical domain, the DermaVQA dataset [29] includes questions, medical history context, one or more images, and detailed responses. Reported BLEU scores for this dataset range from 0.1285 for English, 0.0705 for Chinese, to 0.0135 for Spanish. Unlike the former, our dataset involves responses from medical doctor domain experts answering patient queries. Furthermore, unlike the latter, our dataset specifically focuses on the wound care domain.

3. Methods

In the following sections, we describe the creation of the corpus, baseline systems, and our evaluation methodology.

3.1. Corpus creation

Publicly available consumer health queries were scraped from two websites (tieba.baidu.com and zhidao.baidu.com) where the topic was for wounds. Only posts with images were kept. Three Chinese–English bilingual translators, two trained registered nurses and one medical translator participated in the translation from Chinese to English. Three United States medical doctors, one medical surgeon and one emergency department doctor, and one emergency doctor resident, were tasked to create free response labels to the health queries as if they were answering a real patient question electronically. Responses were then translated back to Chinese.

The dataset was randomly divided into training, validation, and test splits. The training set (279 instances) was single annotated by a single annotator (A1). The validation set (105 instances) had at two different annotators. The test set (93 instances) included responses from all three medical doctors. The final data statistics are shown in Table 2. Mean token counts for the queries and responses are shown in Table 3. To measure whether or not experts agree with each other, we asked one annotator, A1, to review all reference responses and mark those they disagreed with. As a result, the rater only marked 10 for disagreements out of a total of 396 responses from both test and valid sets, when not written by themself.

Additionally, annotators were asked to attach metadata related to each query related to the wound. The wound metadata categories' descriptions and possible label options include:

- Anatomic location (41 classes, e.g., abdomen, fingernail, lower leg, neck, wrist)
- Wound type (8 classes, e.g., pressure, arterial, venous, surgical, diabetic, traumatic)
- Wound thickness (6 classes, e.g., stage I: Superficial, involving only the epidermal layer, stage III: Full thickness, extends through the dermis and into the adipose tissues)
- Tissue Color (6 classes, e.g., pink or pearly white; red and moist; hard, dry and black)
- Drainage Amount (6 classes, e.g., no exudate, minimal amount, moderate amount of drainage)
- Drainage Type (5 classes, e.g., sanguineous, serous, serosanguineous, purulent)
- Infection (3 classes: infected, not infected, unclear)

To gauge agreement for the task, all three annotators were also asked to annotate a small sample of 37 instances. The agreement is shown in Table 4. One annotator, A1, annotated the rest of the instances' metadata. Although individual pairwise categories were in the range of 0.4–0.8, on analysis we found in case of differences, most instances had a majority vote answer. Specifically, for instances where at least one difference occurred, we counted the number of instances where 2 annotators (out of a total 3) agreed. For a high fraction of possible majority votes, this signals that even when there are annotation disagreements, with 3 experts there can be a majority vote conclusion. Furthermore, differences were related to label categories which were semantically close (e.g. stage I versus II, or pink versus red). This suggests that disagreements may be due to small differences in opinions rather than large conceptual differences.

Table 2
Data statistics.

Split	#Instances	#Annotators	#Responses	#Images
Training	279	1	279	449
Validation	105	2	210	147
Test	93	3	279	152

Table 3
Response length statistics (Mean Token Count). English tokens are per word, Chinese per character.

Split	EN		ZH	
	Query	Response	Query	Response
Training	46	29	52	43
Validation	44	41	50	60
Test	52	47	60	68

Table 4
Metadata IAA. Majority gives the fraction of instances where there exists a 2/3 majority answer.

category	A1–A2	A1–A3	A2–A3	majority
anatomic_locations	0.486	0.486	0.811	0.811
wound_type	0.892	0.865	0.919	1.000
wound_thickness	0.405	0.324	0.649	0.892
tissue_color	0.622	0.568	0.757	0.973
drainage_amount	0.514	0.541	0.730	0.865
drainage_type	0.649	0.676	0.838	0.919
infection	0.784	0.703	0.730	0.973

3.2. Baseline systems

We used the following baseline models for the tasks of response generation and image classification:

Gemini-1.5-Pro. We used Google's general AI model gemini-1.5-pro API, with both image and text capabilities, pointing to the "gemini-1.0-pro-002" model. (https://ai.google.dev/gemini-api/docs/models/gemini#gemini-1.5-pro)

GPT-4o. A vision-language-enabled general AI model, part of the greater GPT4 family. We used the Azure OpenAI service API implementation, model version 2024-05-13. Unlike the standard API, safety filters were removed to prevent API filters from denying responses because of image content containing blood or wounds. (https://azure.microsoft.com/en-us/products/ai-services/openai-service)

Qwen-VL-Chat. An instruction-enabled visual language model trained in both English and Chinese from Alibaba Cloud as part of the Qwen model family. We ran this model using the hugging face implementation on a GPU server with 2 x (NVIDIA Tesla V100 16 GB GPU).
(https://huggingface.co/Qwen/Qwen-VL-Chat)
We used the following prompts:

- Response Generation: English: *"Please answer as a professional medical doctor, answer limited to 41 words. {QUERY_TITLE}: {QUERY_CONTENT}"* Chinese: "请以专业医生的身份提供建议，答案只限60字 *{QUERY_TITLE}: {QUERY_CONTENT}* "
  - Length suggestions used the average response lengths in the validation sets.
- Wound Metadata Classification: *You are a skilled medical doctor reviewing medical images (and associated questions) sent by patients. Classify the wound shown in the image into one of these classes. Respond with the exact class name only without additional explanation or details. Classes: [].*
  - The classes options will be filled in "[]" according to the metadata type. Table B.20 describes the used classification labels.

3.3. Evaluation

Automatic Evaluation. Wound image metadata classification evaluation was measured using accuracy. For textual response evaluation, we used three common automatic natural language generation evaluation metrics including BLEU [30], BERTScore [31], and ROUGE-L [32]. Evaluation code is included with the data release.

Human Evaluation. Human evaluation for system textual output was divided according to language. For the English subset, 2 currently practicing medical doctors were recruited and asked to rate system responses along five axes (Disagree Flag,[1] Completeness, Factual Accuracy, Relevancy, Writing Style, Overall) using a three point scale (1.0 accurate and complete answer, 0.5 partially good answer, 0.0 inaccurate or harmful answer). A trained Chinese medical doctor conducted the same study for the Chinese language outputs. We calculate Kendall, Pearson, and Spearman correlations for each rating.

4. Results

We evaluate wound metadata classification results using (i) the images only, (ii) images and English questions, and (iii) images and Chinese questions using GPT-4o, which showed good performance across all settings in initial experiments (Table 5). Adding textual context improves scores for all categories except Tissue Color. Using the English and Chinese context showed similar performances, with English higher for Anatomic Location, Wound Type, Drainage Type, and Infection, and with Chinese higher for Wound Thickness and Drainage Amount.

Table 6 describes results for the response generation task. For both English and Chinese subsets, GPT-4o is ranked highest among two out of three metrics, with Gemini-1.5-Pro ranked highest for the remaining metric (BLEU for English and ROUGE-L for Chinese). However, it is important to note that the score differences between all systems are minimal, with a maximum difference of 0.02 across all metrics and language subsets.

Table 7 presents comparison of system results as judged by a human expert. For the English subset, similar to automatic results, the gap

[1] binary value, 1 for if rater disagrees with response, 0 otherwise

Table 5
Classification results (GPT-4o).

Classification (Accuracy)	Images	Images+EN questions	Images+ZH questions
Anatomic Location	0.5329	**0.6447**	0.6250
Wound Type	0.4671	**0.7763**	0.7697
Wound Thickness	0.3355	0.4145	**0.4210**
Tissue Color	**0.3224**	0.2960	0.2895
Drainage Amount	0.4013	0.4145	**0.4474**
Drainage Type	0.5395	**0.5592**	0.5197
Infection	0.3947	**0.4737**	0.4408

Table 6
Response generation results.

	EN			ZH		
	Gemini	GPT-4o	Qwen-VL	Gemini	GPT-4o	Qwen-VL
BERTScore	0.6556	**0.6640**	0.6348	0.6848	**0.6923**	0.6817
BLEU	**0.0639**	0.0622	0.0510	0.1181	**0.1234**	0.0935
ROUGE-L	0.4493	**0.4501**	0.4282	**0.5006**	0.4959	0.4841

Table 7
Human evaluation of response generation results for test set.

	Gemini	GPT-4o	Qwen-VL
EN			
Disagree Flag (↓)	0.1774	**0.1667**	0.3280
Completeness	**0.9623**	0.9570	0.8468
Factual Accuracy	**0.8817**	0.8763	0.7554
Relevance	**0.9973**	0.9946	0.9381
Writing Style	0.9839	**0.9892**	0.8900
Overall	0.8575	**0.8629**	0.6801
ZH			
Disagree Flag (↓)	0.2043	**0.2151**	0.3871
Completeness	**0.9247**	0.9032	0.8333
Factual Accuracy	**0.8817**	0.8495	0.7634
Relevance	**0.9946**	0.9785	0.9355
Writing Style	**1.000**	0.9892	0.9516
Overall	**0.8226**	0.7957	0.6505

between GPT-4o and Gemini-1.5-Pro is small with 0.01 difference in the overall rating. Unlike automatic metrics, differences between the Qwen-VL-Chat and the other two systems had a larger comparison gap. For the Chinese subset, Gemini-1.5-Pro had the highest results and consistently performed better across all but one category. Differences between Gemini and GPT-4o were slightly larger, at 0.03 difference for the overall metric. The Qwen-VL model likewise had lower scores in the Chinese partition. Consistent with automatic metrics, performance differences were similarly small, at a 0.03 score difference.

We further qualitatively studied a random 20 instances and each of it's 3 gold responses, 3 system responses, and English and Chinese counterparts. We identified several key findings and provide examples in Tables 8, 9, and 10:

- Doctor responses, even for similar responses, will have small divergences in next step-advice and optional cautionary recommendations. As shown Table 9 example, this can lead to divergence in the final recommendation (vaccination versus no vaccination).
- English and Chinese system responses are generated by separate model inference processes, with prompts in its respective language (versus automatically translated). Though in many cases, the English/Chinese response per model (e.g. GPT-4o) are near translations of each other; this is not always the case. For example, one English Gemini output was “These are abrasions (scrapes). Healing time varies, typically 1–3 weeks depending on depth and care. Keep clean, covered, and avoid basketball until pain-free and no risk of re-injury or infection. See a doctor if signs of infection develop”. however the Chinese output was “图片显示擦伤较深, 可能需要2-3周愈合。保持伤口清洁干燥, 使用抗生素软膏预防感染。避免篮球训练直至伤口结痂, 避免再次摩擦受伤部位。请就医以获得专业评估和治疗。” The English mentions 1–3 weeks for healing, while the Chinese output suggests 2–3 weeks. This suggests that though the models have shared knowledge that is language agnostic, still differences in data training lead to divergences in reading in generated content.
- For English outputs, Gemini and GPT-4o models tend to be more confident than the Qwen-VL model. This works in the latter's favor when the case is more ambiguous; though the opposite effect occurs when the case is clear. Table 8 provides an example when the Gemini and GPT-4o model responses gives answers that are more confident than the human domain experts.
- Qwen-VL English outputs tend to have more hedging language then when prompted in Chinese. For example, frequent response pattern includes a prefix of: “It is difficult to determine the healing time without a proper examination by a medical professional” prior to giving an advice. In some cases, it even prefaces answers with a self-disclaimer. For example, the English may have an output: “I'm sorry, but as an AI language model, I am not able to provide medical advice or diagnose conditions. It is important to seek medical attention from a qualified healthcare provider for proper evaluation and treatment of any wound or injury. They will be able to assess the wound, determine the cause, and provide appropriate care” however the corresponding Chinese output would be “您的腿上的伤口可能是由昆虫咬伤或外伤引起的, 建议您清洁伤口并涂抹抗生素软膏。如果症状持续或恶化, 建议您就医检查。” // *The wound on your leg may have been caused by insect bites or trauma. It is recommended that you clean the wound and apply antibiotic ointment. If the symptoms persist or worsen, it is recommended that you seek medical examination.*
- Qwen-VL Chinese outputs may contain mixed language sentences. One example generated mixed sentence is “It is best to keep the

Table 8
Example where gold standard responses have similar answers, but different next steps. Gemini and GPT-4o responses are more confident than human domain experts.

QUERY 各位大神看看这个需要处理不 Can someone take a look to see if this require treatment? 被一个铁边的广告牌撞的，需要缝针或者特别处理吗，以后留疤的几率大不大 I got crash onto an advertisement board that has a metal lining. Do I need suture or special treatment? How likely will it leave a scar
GOLD-RESPONSES Annotator 1: 很难判断是否会留下疤痕。由于伤口位于面部，缝合可能有助于降低留下疤痕的可能性。如果您在过去5年内没有接种过破伤风疫苗，我建议您接种。外出时使用防晒霜或戴帽子，以尽量减少该区域的日晒暴露。 It is difficult to say if it will leave a scar. Since it is located on the face, sutures may help decrease this possibility. I would recommend a tetanus vaccine if you have not had one in the past 5 years. Use sunscreen or wear a hat when outdoors to minimize sun exposure to the area. Annotator 2: 目前无法明确判断是撕裂伤还是仅为擦伤。如果是有组织受损的撕裂伤，建议缝合以减少瘢痕形成。否则局部伤口护理、使用抗生素软膏和无粘连敷料可能就足够了。建议每天1至3次用温肥皂水轻柔清洗伤口。为减少瘢痕形成，建议在一年内避免紫外线照射。 Difficult to fully appreciate if there is any underlying laceration or just an abrasion present. If there is a laceration with tissue gapping, would recommend suture repair for scar minimization. Otherwise local wound care with topical antibiotic ointment and nonadherent dressing is likely sufficient. Gentle cleaning of the wound with warm soapy water 1-3 times daily is recommended. For scar minimization, avoid UV exposure for up to 1 year. Annotator 3: 根据提供的图像很难说。根据受伤的深度，你可能需要缝合或使用粘合剂将伤口边缘粘在一起。它可能会留下疤痕，我建议避免阳光照射到该区域，并涂抹适当的防晒霜以减少疤痕的可能性 It is hard to say based on the image provided, based on the depth of the injury you may either need sutures or a adhesive glue to bring the edges of the wound together. It may leave a scar, I would recommend avoiding sunlight to the area and applying appropriate sunscreen to reduce the likelihood of scarring.
SYSTEM-RESPONSES GEMINI: 建议立即到医院就诊。伤口较浅，可能不需要缝合，但需要专业清创包扎以防止感染。面部伤口留疤可能性较小，但因人而异，建议咨询医生使用祛疤药物。 This laceration appears superficial and may not require sutures. Clean the wound thoroughly with soap and water, then apply antibiotic ointment and a bandage. Scarring is possible, but minimizing tension and sun exposure can help. See a doctor if it worsens. GPT-4O: 建议尽快去医院检查和处理伤口，可能需要缝针。及时治疗和护理可减少留疤风险。 The wound appears to be superficial. Clean it with antiseptic, apply a bandage, and monitor for infection. A visit to a healthcare professional is advisable to assess the need for sutures. Minimal scarring is expected with proper care. QWEN-VL: 从图片上看不出具体的伤势情况，建议尽快就医检查，如果需要缝针或者特殊处理的话应该及时进行，以避免感染和留下疤痕。 It is difficult to determine the extent of the injury without more information. If you have a cut on your forehead that is bleeding, it is a good idea to seek medical attention to ensure that there is no skull fracture or other serious injury. The cut may need to be cleaned and stitched to prevent infection and promote healing. A scar may form over time, depending on the extent of the injury and how well the wound heals.

wound clean and dry, and apply antiseptic cream or ointment. If the wound is deep or won't stop bleeding, it is best to see a doctor for 缝合.". Though remarkably the rendered Chinese is "stitches", which is the correct idea.

5. Discussion

In classification tasks, inter-annotator agreement provides an upper bound on task performance, highlighting performance ceilings which occur when domain experts disagree. Although in language generation tasks additional metric uncertainty exists, measuring one gold response candidate against other gold standard responses with automatic metrics may provide us with comparison. Table 11 presents the result of this experiment. As shown, the scores of gold responses against other gold responses carry similar values to that of the system response. The reference expert score was the proportion of other gold responses that a human annotator judged as acceptable (excluding their own authored response). Moreover the values of the automatic metrics can both over and underestimate the true score given by the expert. This suggests that current metrics cannot adequately account for multiple valid gold references.

In response generation, GPT-4o and Gemini-1.5-Pro showed similar performances when evaluated by human experts. In contrast, the Qwen-VL-Chat model showed lower performance in human evaluation for English compared to its performance in automatic metrics. This discrepancy may be due to the fact that the Qwen-VL-Chat model assessed here is not the largest of the family class. Performance comparisons in Chinese were more closely aligned. The Qwen-VL-Chat model was selected for its open-source nature, instruct-enabled capabilities, and visual language processing, alongside its ability to handle both English and Chinese, at a reasonable performance, in a modest training environment. Although our analysis highlights out-of-the-box performance, this model can be further fine-tuned in future work for improved results.

We further studied the correlation of automatic metrics with human ratings, as presented in Table 12. Human–human (A1/A2) ratings when measured across all evaluation criteria and system outputs, match

Table 9
Example where gold and system responses echo similar information with variations in the amount of emphasis and explanations; though with different final recommendations (no vaccination versus vaccination).

QUERY

被猫隔着厚牛仔裤抓了一下，留下一道血痕，伤口如图，请问需要接种狂犬疫苗吗

I was scratched by a cat through thick jeans, leaving a blood mark as shown in the picture. Do I need to get a rabies vaccine?

GOLD-RESPONSES

Annotator 1: 在这种情况下不需要注射狂犬疫苗。

No need for rabies vaccine in this situation.

Annotator 2: 如果猫的疫苗接种状态未知且无法隔离，虽然风险较低，具体取决于地区，建议接种疫苗。否则，局部伤口护理，使用局部抗生素和敷料是就行了。

If vaccination status of cat unknown and cannot be quarantined, while risk is low, depending on the area, would recommend vaccination. Otherwise, local wound care with topical antibiotic and dressing is sufficient

Annotator 3: 你应该不需要为猫抓伤接种狂犬症疫苗，因为从抓伤传播狂犬症的风险非常低。当然，如果你居住在一个狂犬症发病率很高的也区，我会另作考虑。虽然如此，我会继续观察伤口部位；如果伤口已受感染，你可能需要抗生素来治疗。我会用抗生素软膏包扎伤口。

You should not need to get a rabies vaccine for a cat scratch as the risk of transmission of rabies from a scratch is very low however, if you live in an area with high instance, I would consider it. However, I would continue to keep an eye on the area as you may need antibiotics for the cat scratch if it becomes infected. I would dress the wound with antibiotic ointment.

SYSTEM-RESPONSES

GEMINI: 牛仔裤阻隔了猫的牙齿，皮肤破损轻微，未见穿透真皮的深部伤口。无需接种狂犬疫苗，用肥皂水清洗伤口，并保持清洁干燥即可。如有感染迹象请及时就医。

The jeans likely prevented a true bite. Rabies transmission requires saliva contact with an open wound. Wash the area thoroughly. Rabies vaccination is highly unlikely to be necessary in this situation. Observe the cat for 10 days for any signs of illness. Consult a doctor if you have any concerns.

GPT-4O: 建议尽快用肥皂水清洗伤口并消毒，然后去医院咨询医生。通常，猫抓过建议接种狂犬疫苗以预防感染。

Rabies is rarely transmitted through scratches. Clean the wound thoroughly and monitor for infection. Seek medical attention if the cat's rabies vaccination status is unknown or if the cat is acting abnormally. Your doctor can provide the best advice.

QWEN-VL: 如果猫是家养且接种过狂犬疫苗，且你没有抓伤伤口感染，一般不需要特殊处理，观察几天看看。如果猫是野猫，建议去打狂犬疫苗。

If you were scratched by a cat and it broke the skin, you should get a rabies vaccine. The vaccine can prevent rabies if given before symptoms appear. Contact your healthcare provider or local health department for more information.

accuracies was at 80% (Table A.13). However, in terms of correlations there was only moderate correlation for Disagree Flag, Factual Accuracy, and Overall, at a range of 0.46–0.61, with lower correlations for Completeness, Relevance, and Writing Style. This is likely because, although absolute match accuracy is high, the relationship between A1 and A2 ratings did not follow a clear linear or ranking relationship.

In the English subset, BERTScore was shown to have the highest correlations with human ratings in for a majority of the evaluation axes and correlation measures. In contrast, BLEU was shown to have the highest correlation with human ratings in the Chinese subset for most evaluation axes. BERTScore and Rouge also had better correlations in completeness, relevance, and writing-style. As expected, the correlation values were highly dependent on the method used to fit the correlation. However, across all correlation algorithms, except for Pearson correlation in relevance and writing style, none of the metrics demonstrated a correlation higher than a modest 30%. Unlike other natural language generation tasks, such as machine translation and summarization, response generation in the medical domain involves significantly greater content variation due to differing expert opinions. Natural language evaluation methods for open-responses, especially for the medical domain, still requires much further improvement and study.

6. Conclusion

By combining multimodal data sources such as images, text, and clinical information, the WoundcareVQA dataset aims to promote the creation and evaluation of efficient, context-aware VQA systems that can assist healthcare providers in answering wound care-related questions.

In this study, we observe a challenging landscape. Performance ranges were modest not only for response generation, but also for metadata classification. For response generation, this is consistent with dialog generation scores and open-response long-context VQA tasks. For classification, this was consistent with measured agreements. In this work, we only study out-of-the-box performances for several models. Further improvements may be achieved by experimenting with few-shot prompting, fine-tuning models, data augmentation, or more complex systems. Automatic metrics related to natural language generation evaluation are especially tricky given the unconstrained nature of possible gold standard labels and expert opinion divergences. Modest correlation of automatic metrics to human ratings, prove evaluation to be critical area for future research. Finally, studying the incorporation of this technology in a real healthcare setting, e.g. providing recommended email response drafts, is crucial in future work.

We believe the new WoundcareVQA dataset (https://osf.io/xsj5u/), released with standard train/valid/test splits, will support the development and benchmarking of innovative medical systems for multimodal response generation, aiming to alleviate the burden on healthcare providers and to improve overall healthcare workflows.

Table 10
Example where gold standard responses universally agree there is no infections, however system responses recommend urgent medical attention.

QUERY 骑车摔伤打了6次屁股消炎针。天天吃消炎药。 I got six shots of anti-biotics on my butt since I was fallen from the scooter, and I have to take medicine everyday. 14天了还是这样子。该怎么办。 What should I do? It has been forteen days but no heal.
GOLD-RESPONSES Annotator 1: 伤口可能需要长达8周才能愈合，但目前没有感染迹象。请继续进行伤口护理，每天用肥皂和水清洗伤口，并用绷带包扎。 It may take up to 8 weeks for healing but there is no sign of infection at this time. Continue performing wound care with washing the area with soap and water daily and covering with a bandage. Annotator 2: 无明显的继发感染迹象。伤口表现为溃疡，并有部分肉芽组织。由于伤口位于足部，预计愈合时间较长。为促进愈合，避免使用酒精、双氧水或碘等干燥剂。保持伤口覆盖无粘连敷料，并在伤口上涂抹抗生素软膏，可每日更换敷料。 No overt signs of secondary infection. Wound appears to be ulcerated with some granulation tissue present. Given that it appears to be on the foot it would likely require a lengthy period of time for healing. To promote healing, avoid any drying agents such as alcohol, hydroperoxide, or iodine. Keep the area covered with a nonadherent dressing and apply topical antibiotic ointment to the wound. You may change the dressing daily. Annotator 3: 伤口现在似乎痊愈得很好，这需要一些时间才能完在愈合。我建议将伤口浸泡在温肥皂水中，以帮助预防感染。我会用杆菌肽软膏和XEROFORM敷料包扎伤口。 This wound appears to be healing well at this time, this will take some time to heal. I would recommend soaking the area in warm soapy water to help prevent infection. I would dress the wound with bacitracin ointment and xeroform dressing.
SYSTEM-RESPONSES GEMINI: 伤口感染未愈，两周消炎治疗无效需立即就医。请前往医院进行伤口清创、评估感染情况，并根据具体情况调整抗生素治疗方案，必要时进行细菌培养。避免自行处理。 This wound requires immediate medical attention. It appears infected and needs professional evaluation and treatment. Please see a doctor or go to an urgent care facility as soon as possible. Do not delay. GPT-4O: 建议尽快就医，可能需要进一步检查和治疗，包括清创、换药以及可能的抗生素调整。避免受伤部位继续受压，保持清洁干燥。 Please see a doctor urgently. You may need wound re-evaluation, potential debridement, and possibly a change in antibiotics, as a non-healing wound can lead to serious complications. QWEN-VL: 您的情况可能需要进一步的医疗处理，建议尽快就医。在就医前，可以先暂停使用消炎药，并观察伤口的变化。 It's possible that the infection has spread to other areas of your body or that you have developed an antibiotic-resistant infection. I would recommend that you see a doctor or a podiatrist for an evaluation and treatment. They may need to prescribe a different antibiotic or perform a procedure to remove the infected tissue.

Table 11
Evaluation results for the WoundCareVQA Holdout-One-Gold experiment. Each row compares (i) a gold response evaluated against two other gold references (*Gold-Holdout*) and (ii) a GPT-4o response evaluated against all references.

	Gold-Holdout	GPT-4o
EN		
BLEU	0.0469	0.0622
BertScore	0.6634	0.6640
ROUGE-L	0.4353	0.4499
Human-Expert	0.975	0.8629
ZH		
BLEU	0.1109	0.1234
BertScore	0.6896	0.6923
ROUGE-L	0.2254	0.4961
Human-Expert	0.975	0.7832

Table 12
Correlations between human and automatic evaluation of response generation for test set. A1/A2 shows correlations between two English domain expert raters (the higher, the better). For Disagree Flag correlation with automatic metrics, the more negative the better.

	EN			ZH		
	kendall	pearson	spearman	kendall	pearson	spearman
completeness						
bertscore	**0.1705**	**0.2378**	**0.2092**	0.1050	**0.1582**	0.1283
bleu	0.1003	0.1098	0.1233	0.0474	0.0754	0.0586
rouge	0.1373	0.2373	0.1685	**0.1312**	0.1540	**0.1599**
A1/A2	0.2296	0.2080	0.2331	–	–	–
disagree_flag						
bertscore	−0.1364	**−0.1675**	−0.1667	−0.0106	−0.0176	−0.0129
bleu	**−0.1483**	−0.1367	**−0.1813**	**−0.0912**	**−0.0973**	**−0.1115**
rouge	−0.0897	−0.153	−0.1096	0.0217	0.0141	0.0265
A1/A2	0.4636	0.4636	0.4636	–	–	–
factual-accuracy						
bertscore	**0.1101**	**0.1872**	**0.1377**	0.0922	0.1639	0.1134
bleu	0.0897	0.0989	0.1124	**0.1534**	**0.1867**	**0.1925**
rouge	0.0262	0.1717	0.0326	0.0132	0.0307	0.0167
A1/A2	0.4782	0.5589	0.4973	–	–	–
relevance						
bertscore	**0.1339**	0.2294	**0.1642**	**0.1035**	**0.1636**	**0.1271**
bleu	0.1031	0.115	0.1263	0.0663	0.0769	0.0815
rouge	0.1164	**0.3288**	0.1426	0.0406	0.0427	0.049
A1/A2	0.3047	0.3123	0.3063	–	–	–
writing-style						
bertscore	**0.1485**	0.2596	**0.1822**	0.0138	0.0188	0.0169
bleu	0.1247	0.1502	0.1534	0.0628	0.0773	0.0768
rouge	0.0852	**0.3181**	0.1047	**0.0742**	**0.0904**	**0.0907**
A1/A2	0.3830	0.5473	0.3867	–	–	–
overall						
bertscore	**0.1782**	**0.2515**	**0.2227**	0.1589	0.2240	0.1963
bleu	0.1605	0.1687	0.2007	**0.1792**	**0.2226**	**0.2252**
rouge	0.0993	0.2193	0.1247	0.0539	0.0612	0.0682
A1/A2	0.5550	0.6114	0.5812	–	–	–

Table A.13
A1/A2 response generation ratings agreement for test set.

Category	Accuracy	RMSE
Disagree Flag	0.810	0.190
Completeness	0.789	0.074
Factual Accuracy	0.713	0.085
Relevance	0.939	0.023
Writing Style	0.914	0.030
Overall	0.674	0.098
ALL	0.806	0.083

Table A.14
A1/A2 disagree flag rating confusion matrix.

	0.0	0.5	1.0
0.0	190	0	18
0.5	0	0	0
1.0	35	0	36

CRediT authorship contribution statement

Wen-wai Yim: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Asma Ben Abacha:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Robert Doerning:** Writing – review & editing, Data curation. **Chia-Yu Chen:** Data curation. **Jiaying Xu:** Data curation. **Anita Subbarao:** Data curation. **Zixuan Yu:** Data curation. **Fei Xia:** Writing – review & editing, Project administration, Methodology. **M. Kennedy Hall:** Writing – review & editing, Project administration. **Meliha Yetisgen:** Writing – review & editing, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Response generation human evaluation agreements and confusion matrices for test set

See Tables A.13–A.19.

Appendix B. Wound metadata classification labels

See Table B.20.

Table A.15
A1/A2 completeness rating confusion matrix.

	0.0	0.5	1.0
0.0	0	0	1
0.5	2	8	6
1.0	7	43	212

Table A.16
A1/A2 factual accuracy rating confusion matrix.

	0.0	0.5	1.0
0.0	7	21	4
0.5	1	12	23
1.0	1	30	180

Table A.17
A1/A2 relevance rating confusion matrix.

	0.0	0.5	1.0
0.0	0	0	0
0.5	1	2	2
1.0	3	11	260

Table A.18
A1/A2 writing style rating confusion matrix.

	0.0	0.5	1.0
0.0	4	0	0
0.5	1	3	9
1.0	3	11	248

Table A.19
A1/A2 overall rating confusion matrix.

	0.0	0.5	1.0
0.0	5	30	6
0.5	1	22	37
1.0	0	17	161

Table B.20
Wound metadata classification labels..

Task	Used labels [Output classes]
Anatomic Location	"abdomen", "ankle", "arm", "armpit", "back", "backofhead", "chest", "chin", "ear", "elbow", "eyeregion", "face", "fingernail", "fingers", "fingers-interdigital", "foot", "foot-sole", "forearm", "forehead", "groin", "hand", "hand-back", "heel", "knee", "lips", "lowerback", "lowerleg", "mouth", "napeofneck", "neck", "nose", "palm", "scalp", "shoulder", "thigh", "toenail", "toes", "toes-interdigital", "tongue", "wrist", "unidentifiable"
Wound Type	"pressure wound usually over bony prominence" ["pressure"], "arterial wound" ["arterial"], "venous wound" ["venous"], "post surgical incisions, flaps, grafts" ["surgical"], "diabetic wound" ["diabetic"], "traumatic wound includes skin tears, burns, traumatic injuries" ["traumatic"], "atypical wound, bullous pemphigoid, pyoderma gangrenosum" ["atypical"], "unknown wound" ["unknown"]
Wound Thickness	"superficial wound, involving only the epidermal layer" ["stage_I"], "partial wound thickness affects the epidermis and may extend into the dermis" ["stage_II"], "full wound thickness, extends through the dermis and into the adipose tissues" ["stage_III"], "full wound thickness, extends through the dermis and adipose, exposing muscle, bone, fascia, or tendon" ["stage_IV"], "full-thickness skin and tissue loss obscured by slough or eschar" ["unstageable"], "not applicable" ["not_applicable"]
Tissue Color	"pink white tissue color" ["pink_white"], "red moist tissue color" ["red_moist"], "yellow brown grey tissue color" ["yellow_brown_grey"], "hard dry black tissue color" ["hard_dry_black"], "red granular tissue color" ["red_granular"], "other tissue color" ["other"]
Drainage Amount	"No exudate" ["no_exudate"], "Scant amount of exudate" ["scant"], "Minimal amount of exudate" ["minimal"], "Moderate amount of drainage" ["moderate"], "Large or copious amount of drainage" ["copious"], "not applicable" ["not_applicable"]
Drainage Type	"sanguineous exudate is fresh bleeding" ["sanguineous"], "serous drainage is clear, thin, watery plasma" ["serous"], "Serosanguineous exudate contains serous drainage with small amounts of blood present" ["serosanguinous"], "Purulent exudate is thick and opaque. It can be tan, yellow, green, or brown in color" ["purulent"], "not applicable" ["not_applicable"]
Infection	"infected", "not infected" ["not_infected"], "unclear"

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